1 A beaker contains water and some metal blocks as shown in Fig. 1.1.

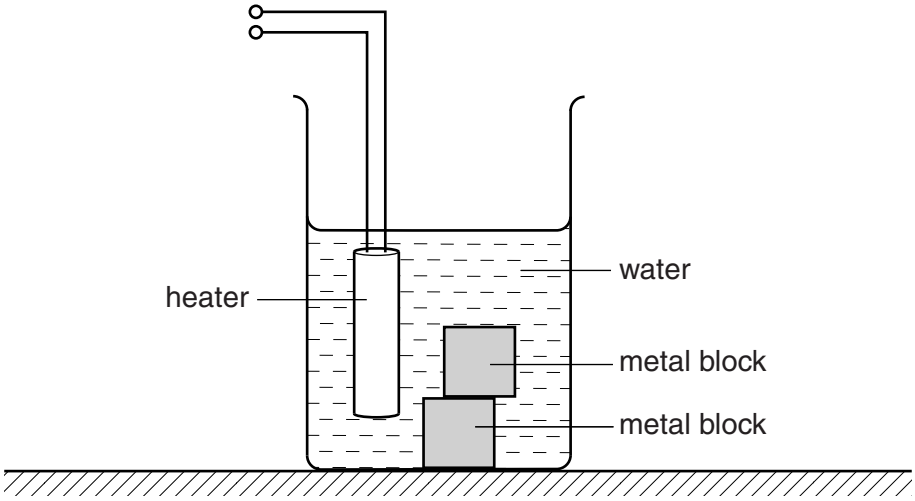

Fig. 1.1

A student uses an electrical heater to produce a particular temperature increase in the water.

It is suggested that the electrical energy E supplied to the heater is related to the mass m of metal blocks by the relationship

$$E = am + b$$

where a and b are constants.

Design a laboratory experiment to test the relationship between E and m. Explain how your results could be used to determine values for a and b. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) the analysis of the data,

(e) the safety precautions to be taken.

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